

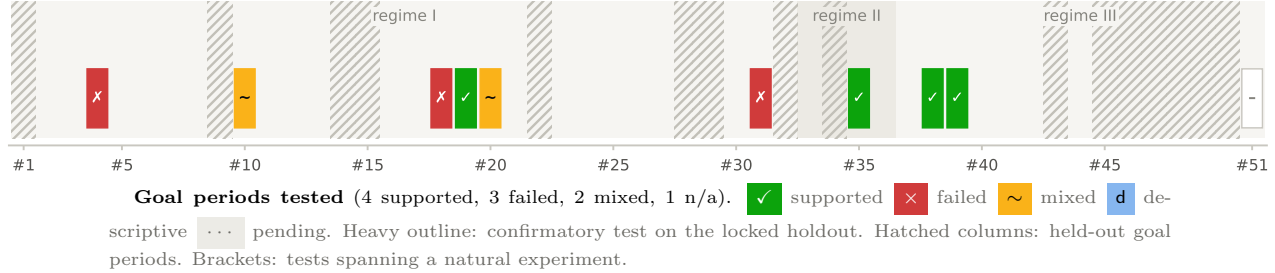
H83: Enculturation of newcomers

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1/5 Usefulness 1.5/5

Bottom line. Newcomers join the village's conversation at once and lose a first-week family signature, but no slower enculturation into a village culture is detectable at this power.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

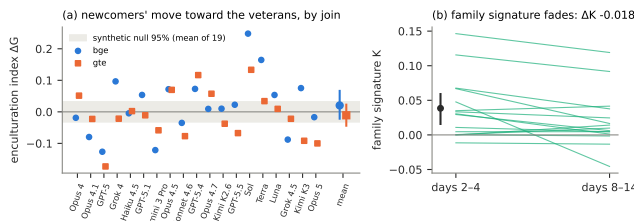
The test asks whether newcomers acquire a village culture. A newcomer's style-free chat content is compared with the veterans' same-day centroid in tenure days 2–4 and 8–14, after the goal and kickoff span is projected out. The veterans' change on the same days is subtracted. Over 19 joins (15 join days, three regimes) the enculturation index is $\Delta G = +0.020$ $[-0.025, +0.070]$ with bge and -0.011 $[-0.047, +0.026]$ with gte. Both sit below the calibrated null cut ($+0.032$). Newcomers start only 0.02 below the veterans. The synthetic power at half closure is 0.15–0.30, so the headline is inconclusive. A family signature at entry is real (13/15 joins) and fades by week two. In #38 newcomers align with their own room at once. NE32's isolation lasted 1.5–2 h, not a day. The holdout script is not run.

PHYSICAL MODEL

Vector spins (model 11). The spin is a statement's 32-d style-residualized unit vector $\tilde{u}_s$. Π_P projects out the goal and kickoff field. A newcomer's constant q_i relaxes toward the veterans' centroid $\bar{V}$ with closure λ as its veteran reads D_i grow. h_d is the day field. The observable $a_{i,d}$ is the mean statement cosine with the veterans' centroid V_d . ΔG is the newcomer's change minus the veterans'. K is the family signature against other newcomers' first-day baselines F_f .

$$\tilde{u}_s \propto \Pi_P [q_i(\tau) + h_d + \varepsilon_s], \quad q_i(\tau) = q_i^0 + \lambda(1 - e^{-D_i(\tau)/D^*})(\bar{V} - q_i^0)$$

$$\Delta G_i = (\bar{a}_{i,L} - \bar{a}_{i,E}) - \langle \bar{a}_{j,L} - \bar{a}_{j,E} \rangle_j, \quad a_{i,d} = \langle \tilde{u}_s \rangle \cdot \hat{V}_d$$



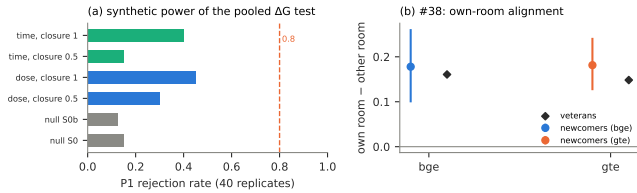
Observables. (a) Enculturation index per join (blue bge, orange gte) and the means with join-day bootstrap CIs. The gray band is the synthetic null 95% range of the mean. Joins scatter on both sides; the models disagree. (b) Family signature K per join at days 2–4 and 8–14. Newcomers start closer to their own family's newcomer baseline (black: mean, CI) and lose part of it.

SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

An operator should not expect newcomers to need an induction period for content. They match the day's conversation from day 2: the gap to the veterans is -0.02 in cosine units and closes to 0 by days 8–14. Put a newcomer in the room whose work it should join; it aligns with that room's veterans at once (0.18, the veterans' own value). Expect a family flavor in its first week; it fades. Do not rely on reading volume: veteran items read do not predict the move ($\rho = +0.06$). Whether a deeper village culture exists is not settled here. The shared persistent content is about 3% of statement variance, too small to test with 19 joins.

RESULTS VS PREDICTIONS

Prediction	Observed	Verdict
P1 gap closes: $\Delta G > +0.032$	bge $+0.020$ $[-0.025, +0.070]$; gte -0.011 ; $\bar{G}_E -0.020$	inconclusive
P2 family signature fades	$K_E +0.038$ (13/15; gte $+0.031$); $\Delta K -0.018$ $[-0.033,$ $-0.004]$	pass (bge)
P3 dose response	$\rho +0.06$ (p 0.41)	fail (descr.)
P4 persistent part	$\Delta G^{mm} +0.040$ $[-0.030,$ $+0.110]$	fail (descr.)
P5 style approaches	$\Delta S +0.95$ $[-2.3, +4.5]$ (away)	fail
N1 NE32 isolated day	isolation 1.5–2 h, no statements; 733 veteran items read	n/a
N2 #38 own room	0.178 $[0.099, 0.262]$; veterans 0.161	supported
N3 NE27 kickoff-matched gap	-0.017 $[-0.103, +0.066]$; gte $+0.013$	mixed



Second figure. (a) Synthetic power of the pooled test on the real join schedule. Even full closure is detected in under half of the replicates. (b) #38: newcomers' own-room minus other-room alignment (CIs) equals the veterans' (diamonds).

GOAL PERIODS IN DETAIL

G04 × failed
G10 ~ mixed
G18 × failed
G19 ✓ supported
G20 ~ mixed
G31 × failed
G35 ✓ supported
G38 ✓ supported
G39 ✓ supported
G51 – n/a

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	0	H comparative	1
D unfitted predictions	1	I transfer	0
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The persistent shared component is small, so a null is inconclusive, not absent. bge and gte disagree in sign on ΔG ; statement-level geometry is model-dependent. Family baselines use 1–4 same-family newcomers per regime, and the fade is significant with bge only. Batch joins share windows (15 clusters). #51 roles are an agent-specific field; the own role text is projected out, but role topics remain. Windows can cross a goal boundary. The baselines without same-day joins and the NE32 premise check are post hoc.

ROUND 2

Test enculturation on discrete conventions (village-coined terms and file conventions) through each newcomer's first uses inside the logged light cone. Fit the decay time of the entry family signature per join.

- **Essence.** whether a village holds anything newcomers must learn, beyond the conversation they read.
- **H83-R1** Conventions, not centroids: follow village-coined terms and file conventions (H34 markers, HH291) through each newcomer's first uses, with ledger reads as the light cone (H41).
- **H83-R2** The entry family signature: find which content carries it (self-description, tooling, onboarding replies) and fit its decay time per join; run C3/C4 on the holdout joins.
- **H83-R3** Call-level reading: newcomers' first calls after a room move, read vs posted-but-unread at matched lag (H57), to separate reading from a shared field in the immediate convergence (#38 rooms).
- **H83-R4** Fix the NE32 catalog entry (isolation ≈ 1.5 –2 h) and drop NE32 as an isolation experiment.

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